Student Performance Analysis

Portfolio Project 2 • Data Analysis & Visualization

Executive Summary

This project analyzes the academic performance of 2,000 simulated students using Python, Pandas, NumPy, Matplotlib, and Seaborn. The analysis covers subject performance, pass/fail outcomes, attendance, study hours, test preparation, gender, parental education, descriptive statistics, and correlations.

The validated dataset contains no missing values or duplicate rows, and all 2,000 Student IDs are unique. The overall pass rate is 48.70%, while the fail rate is 51.30%. Students who completed test preparation had a higher average score and pass rate than students who did not.

Key Performance Indicators

KPI	Value
Students	2,000
Average Math Score	48.46
Average Reading Score	50.48
Average Writing Score	50.59
Average Overall Score	49.84
Pass Rate	48.70%
Fail Rate	51.30%

Subject Performance

Subject	Mean	Median
Math	48.46	48.57
Reading	50.48	50.41
Writing	50.59	50.73
Overall	49.84	49.75

Writing has the highest average score, while Math has the lowest average score.

Test Preparation Analysis

Test Preparation	Average Score	Pass Rate	Students
Completed	52.99	64.50%	724
None	48.06	39.73%	1,276

Students who completed test preparation had an average score about 4.93 points higher and a pass rate about 24.77 percentage points higher than students who did not. This is an observed association in the simulated dataset, not proof of causation.

Attendance, Study Hours, and Correlation

Attendance correlated 0.37 with Overall Score, while Study Hours correlated 0.40. Math, Reading, and Writing correlated 0.68, 0.64, and 0.69 respectively with Overall Score.

Demographic Analysis

Female students averaged 49.77 with a 49.21% pass rate; male students averaged 49.92 with a 48.17% pass rate. The difference is small. Parental education groups also showed only small differences, ranging from approximately 49.49 to 50.39.

Recommendations

• Provide additional Math practice and targeted support.

• Identify students with lower attendance for early intervention.

• Encourage consistent study schedules.

• Promote structured test preparation and revision.

• Use early intervention for students at risk of failing.

• Focus support on individual learning needs.

• Monitor performance, attendance, and study habits regularly.

Conclusion

The project demonstrates how data analysis and visualization can identify academic performance patterns and support practical recommendations. It includes data validation, descriptive statistics, categorical analysis, correlation analysis, and portfolio-ready visualizations. Because the dataset is simulated, findings are intended for analytical demonstration rather than real-world educational decisions.